When and Where You Compute Is What You Emit

A Carbon-Aware MLOps Pipeline for Sustainable AI Infrastructure

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Abstract

Data centers are on pace to become one of the fastest-growing sources of global electricity demand, with AI compute alone projected to push consumption past 800 TWh by 2028, on the order of a mid-sized country's annual draw (IEA, 2025). This paper presents a carbon-aware MLOps pipeline that treats *when* and *where* a workload runs as a decision variable, rather than a fixed constraint, and operationalizes that decision as a single actionable signal: a Green Window Probability and a RUN WORKLOAD / WAIT / ADJUST recommendation. Three machine learning components were designed, trained, and deployed behind a unified inference layer, the Carbon-Opportunity Signal: an XGBoost classifier that labels a forecast window as low- or high-carbon (86.1% accuracy, F1 = 0.87, meeting the project's target), an XGBoost regressor that estimates national data-center power-capacity growth from AI-adoption indicators, and a K-Means (k = 6) clustering step that assigns countries to carbon-sustainability archetypes. The pipeline survived a real-world MLOps failure mode mid-build: two of six proposed datasets were deprecated after the proposal was approved, and the substitution is documented rather than hidden. The regression component fell short of its target (country-grouped cross-validated $R^2 = 0.16$ against a target of 0.70), and that result is reported honestly alongside the leaky, over-optimistic number a naive random split would have produced ($R^2 = 0.51$), because the gap between the two is itself the paper's clearest lesson in MLOps evaluation discipline. The complete signal is served through a local FastAPI development server and an Azure ML Managed Online Endpoint, closing the loop from raw grid data to a deployable, explainable scheduling recommendation.

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1. Introduction and Motivation

1.1 The Hidden Cost of AI Compute

Every inference request, training run, and batch job carries an invisible carbon cost that is a function of two variables the requester almost never controls or even sees: *when* the workload runs, and *where* on the electrical grid it runs. A GPU cluster drawing power from a region running 80% coal at 6 PM on a July weekday emits several times more CO_2 per kilowatt-hour than the same cluster running at 2 AM in a region with high wind penetration. As AI workloads scale, this gap compounds: the International Energy Agency projects that global AI data-center electricity consumption will surpass 800 TWh by 2028, a demand level comparable to the annual consumption of France (IEA, 2025). Most of that growth is being absorbed by existing or newly-built infrastructure with little regard for the carbon intensity of the grid it draws from at the moment it draws it.

This project treats that gap as an MLOps problem rather than a policy problem. Rather than asking organizations to build new low-carbon data centers, a capital-intensive, multi-year undertaking, it asks a narrower, immediately actionable question: given the compute infrastructure that already exists, can a model learn to identify low-carbon windows and locations well enough to shift discretionary workloads toward them, and can that model be operationalized as a production service rather than a one-off notebook analysis?

1.2 Research Question and Approach

Research question: How might we accurately identify low-carbon compute windows and locations across the electrical grid, and operationalize those predictions to support carbon-aware workload scheduling? The approach taken here decomposes the question into three supervised and unsupervised learning problems that are trained independently but combined at inference time into one recommendation:

- **Classification:** is a given region, at a given hour, day-of-week, and month, a low-carbon (“green”) or high-carbon (“red”) compute window?
- **Regression:** what national data-center power-capacity growth does a country's AI-adoption trajectory imply, to prioritize which regions carbon-aware scheduling matters most for?
- **Clustering:** which carbon-sustainability archetype does a country or facility belong to, to add locational context beyond the time-of-day signal alone?

The three outputs are fused into a single artifact, the **Carbon-Opportunity Signal**: a lightweight inference layer that returns a Green Window Probability and a plain-language recommendation, RUN WORKLOAD, WAIT, or ADJUST, that a human operator or an automated orchestrator can act on directly. Section 3 details the system architecture; Sections 5 and 6 walk through algorithm selection and results for each of the three models; Section 7 describes how the outputs are fused; and Section 8 covers deployment to Azure.

2. Data Landscape

The pipeline draws on four data sources spanning three levels of granularity: an hourly, US-grid-region-level electricity monitor; a country-year panel of AI-adoption and infrastructure indicators; a facility-level inventory of US data centers; and a country-year panel of long-run sustainable-energy indicators.

- **EIA-930 Hourly Electric Grid Monitor** (U.S. Energy Information Administration): hourly net generation by fuel source and hourly CO_2 emissions intensity, generated and consumed, across 12 US grid balancing regions. 379,000 rows, 2023 through 2026. This is the classifier's primary training table.
- **AI Readiness and Growth, 150 Countries** (GAID compilation, Kaggle): country-level AI Index sub-scores (Infrastructure, Talent, Commercial, Development, Government Strategy, Operating Environment, Intensity, Research, Scale) alongside National AI Cluster Power Capacity (MW), the regression target.
- **ICARUS-DC** (Johns Hopkins APL / NSF): 354 U.S. data-center facilities with operator, state, and power capacity, mapped to an EIA-930 grid region to support facility-level recommendations.
- **Global Data on Sustainable Energy** (Kaggle): country-year renewable share, low-carbon electricity percentage, and CO_2 emissions, used as clustering features alongside AI infrastructure indicators.

3. System Architecture

The pipeline is organized as four stages, each a discrete, independently re-runnable script: ingestion, training, signal computation, and serving. Figure 1 shows the full data flow from raw downloads through the two serving paths, a local FastAPI development server and an Azure ML Managed Online Endpoint, both of which execute the identical carbon_signal.py inference code; only the hosting layer changes between development and production.

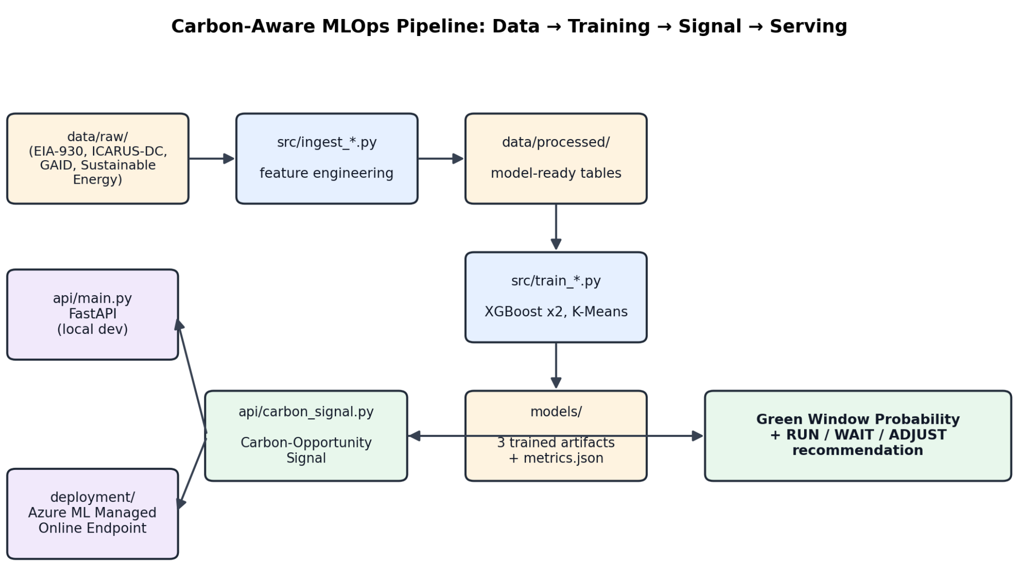

Figure 1. End-to-end pipeline: raw sources are ingested into model-ready tables, three models are trained independently, and the Carbon-Opportunity Signal layer combines their outputs into a single recommendation served identically from FastAPI or Azure ML.

Raw downloads are intentionally excluded from version control (data/raw/ is gitignored) and regenerated on demand by the four ingest_*.py scripts; only the processed, model-ready tables in data/processed/ and the trained artifacts in models/ (plus their metrics JSON) are treated as build outputs worth persisting. This mirrors standard MLOps practice of separating raw data acquisition, which is slow, external, and subject to source drift, from the fast, deterministic feature-engineering and training steps that consume it.

4. Data Engineering: A Real-World Deprecation Challenge

Six datasets were cited and approved in the Preliminary Proposal and Algorithm Selection deliverables (both graded 100/100, July 14, 2026). By the time implementation began, two of the six had been removed from Kaggle entirely, and a third, Electricity Maps' hourly carbon-intensity CSVs, the dataset the proposal explicitly called "the operationally differentiating dataset," had moved its hourly data behind a paid API and account requirement. Silent dataset deprecation mid-project is a legitimate and under-discussed MLOps failure mode: pipelines built against a specific external source are only as durable as that source's availability guarantees, and a production system needs a documented substitution path rather than a broken build. Table 1 documents each substitution rather than quietly absorbing it.

#	Proposal source	Status at build time	Used instead
1	Kaggle: Global Data Center Dataset	404, removed from Kaggle	AI Readiness & Growth (150 Countries): National AI Cluster Power Capacity (MW)
2	Kaggle: Global AI & Data Consumption 2025	404, removed from Kaggle	Same GAID compilation, nine AI Index sub-scores as adoption features
3	Electricity Maps hourly carbon intensity	Free tier removed; now paid API/account	EIA-930 Hourly Grid Monitor: free, public, U.S. government, no login
4	ICARUS-DC (GitHub)	Available	Used as cited
5	LBNL 2024 U.S. Data Center Energy Usage Report	Site blocks automated fetches	Cited for narrative grounding only (never a raw modeling input)
6	Global Data on Sustainable Energy (Kaggle)	Available	Used as cited

The EIA-930 substitution turned out to be a strict upgrade rather than a compromise: it is authoritative (official U.S. government data), free with no registration, and geographically aligned with the ICARUS-DC facility data, since both are U.S.-grid-centric, arguably a cleaner join than the originally proposed Electricity Maps plus ICARUS-DC pairing would have produced. Likewise, the AI Readiness and Growth compilation turned out to cover both missing roles, the data-center-capacity target and the country-level AI-adoption features, in a single source, which is a cleaner join than stitching together two separately-sourced Kaggle datasets would have been.

5. Algorithm Selection and Methodology

Each of the three deliverables was assigned a distinct algorithm based on the structure of its underlying data and the operational requirement that every model run efficiently behind a low-latency Azure Managed Endpoint, support periodic retraining, and expose feature importance so the run/wait/adjust recommendation can be explained rather than treated as a black box.

5.1 Classifier: XGBoost

The Green Window classifier uses gradient-boosted trees (XGBoost) to label a forecast window as green or red from hourly grid carbon intensity, renewable percentage, region one-hot encoding, demand z-score, and cyclical (sine/cosine) encodings of hour-of-day, day-of-week, and month. Boosted trees natively capture the interacting, cyclical features that drive carbon intensity, time-of-day peak demand interacting with renewable availability, and output calibrated class probabilities, which the pipeline needs to produce a graded recommendation rather than a hard label. Logistic regression was retained only as an interpretability baseline; it cannot model the nonlinear hour times renewable-availability interaction central to the research question. A deep neural network was rejected: after country/region-level aggregation the assembled tables are modest in size, insufficient to train a deep model without severe overfitting, and a boosted-tree model is materially cheaper to serve at low latency from a managed endpoint. Best hyperparameters (5-fold grid search): 400 estimators, max depth 6, learning rate 0.1, subsample 0.7, colsample_bytree 0.8, min_child_weight 5.

5.2 Regressor: XGBoost

The capacity-growth regressor predicts national data-center power capacity (MW, log-transformed) from AI-adoption features, principally the AI Index overall score, the scale sub-score, and survey year. The same reasoning that favored boosted trees for classification applies here: capacity growth is unlikely to scale linearly with AI-adoption scores (diminishing or accelerating returns are plausible at the high end), and the panel is small, tabular, and assembled from multiple joined sources with mixed types, conditions XGBoost handles natively without extensive preprocessing.

5.3 Clustering: K-Means (k = 6)

K-Means was retained from the proposal to assign countries to carbon-sustainability archetypes using renewable share, low-carbon electricity percentage, CO_2 emissions, energy intensity, log data-center capacity, and AI infrastructure score. K-Means was preferred over hierarchical clustering and DBSCAN because the archetype features are continuous and approximately bounded after scaling, a six-archetype target maps naturally onto a business-interpretable, fixed k, and the algorithm scales cleanly and predictably as an automatically retrained pipeline step, a property hierarchical clustering and DBSCAN do not share as k grows or shrinks with new incoming data. The trade-off accepted was the assumption of roughly spherical, similarly-sized clusters and sensitivity to feature scale and outliers, both of which require standardizing inputs before fitting and multiple k-means++ restarts for initialization stability.

6. Experimental Results

6.1 Classification: Green Window Probability

The classifier was evaluated on a 75,798-row held-out test split (of 378,986 total rows) and comfortably met both proposal targets: **86.1% accuracy** against a target of at least 85%, and **F1 = 0.87** against a target of at least 0.80. ROC-AUC reached 0.94. Figure 2 shows the confusion matrix; the model's error is roughly balanced between false reds (4,531 green windows misclassified as red, a missed opportunity cost) and false greens (5,990 red windows misclassified as green, the more operationally costly error since it would schedule a workload into a high-carbon window under a false green light).

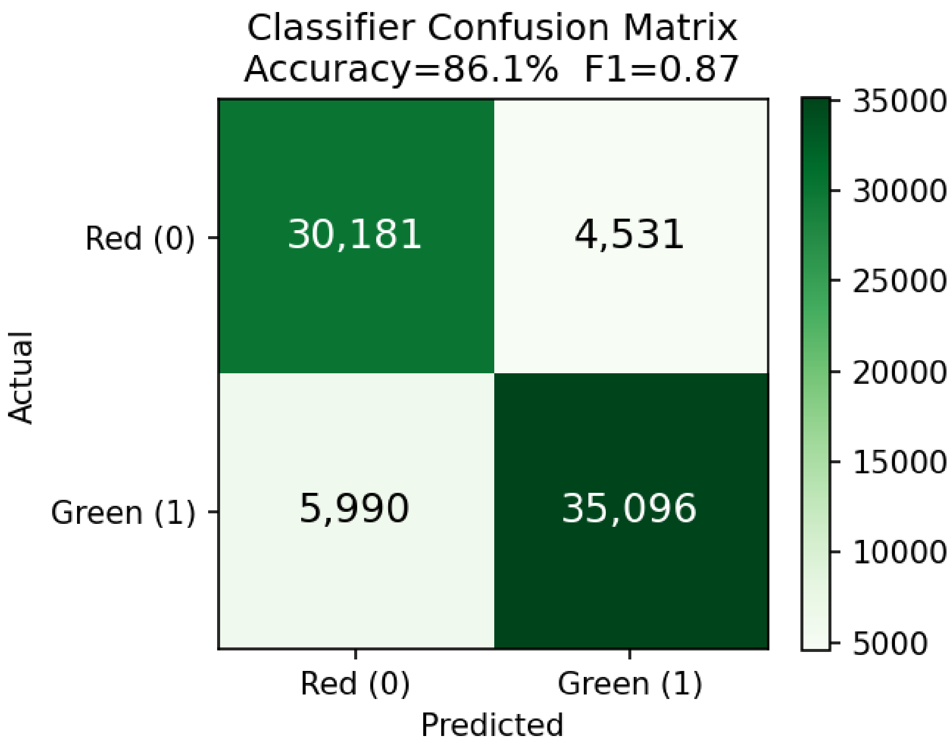

Figure 2. Classifier confusion matrix on the 75,798-row held-out test set. Accuracy 86.1%, F1 0.87, both proposal targets met.

Figure 3 shows the top feature importances. Region indicators dominate: the Northwest (region_NW), Central (region_CENT), and California (region_CAL) one-hot features are the three strongest predictors, ahead of renewable_pct itself. This is an interpretable and, in retrospect, unsurprising result: which grid you are on is a stronger signal than the instantaneous renewable percentage alone, because each region's baseline fuel mix (hydro-heavy Northwest versus coal-heavier Central) sets a very different carbon-intensity floor and ceiling before time-of-day effects are even considered. Per-region green/red thresholds (kg CO_2/kWh) were fit individually, ranging from 0.4 in California to 1.1 in the Midwest, rather than applying one global cutoff, precisely because of this baseline gap.

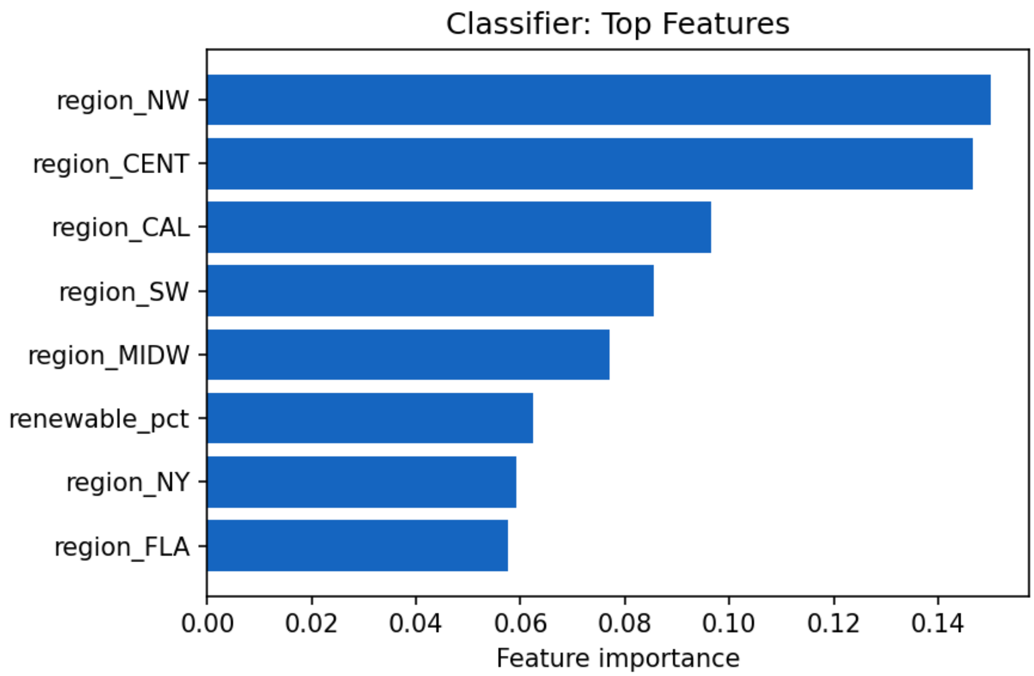

Figure 3. Top eight classifier feature importances. Which grid region a facility sits on outweighs the instantaneous renewable percentage as a predictor of window carbon intensity.

6.2 Regression: National Infrastructure Capacity, and the Leakage Lesson

The regression target, national data-center power capacity (MW) from AI-adoption features across 34 countries, 126 country-year rows, did not meet its target, and that result is reported without qualification: **country-grouped cross-validated R^2 = 0.16** against a target of at least 0.70. What makes this result worth a dedicated figure rather than a footnote is *why* it fell short, and what a naive evaluation would have hidden. A random 80/20 train-test split on this panel, which places different *years* of the same country in both the train and test sets, produces R^2 = 0.51, because the model can partially memorize a country's identity from correlated features seen in another year of that same country's data rather than genuinely learning the AI-adoption to capacity relationship. A country-grouped 5-fold split, where every row for a given country is held entirely out of training in at least one fold, removes that leakage and drops the honest estimate to R^2 = 0.16 (with three of five folds achieving *negative* R^2, meaning the model performs worse than predicting the mean for held-out countries). Figure 4 makes the three numbers, leaky split, honest split, and target, visible side by side.

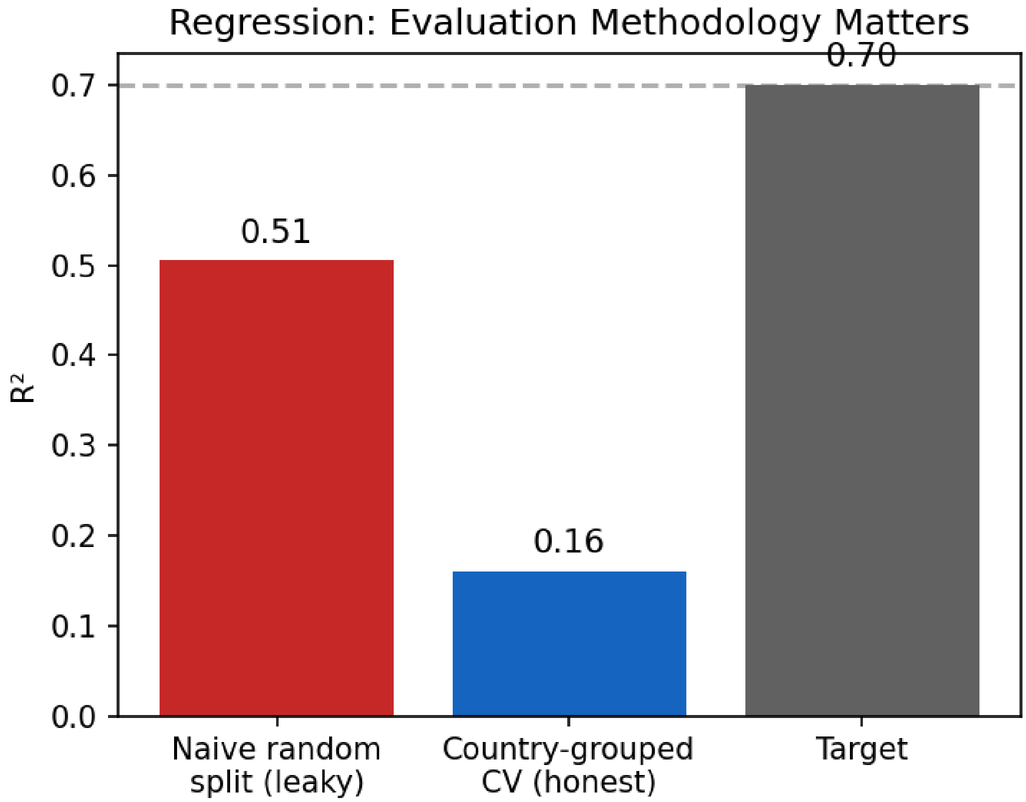

Figure 4. Evaluation methodology matters more than the model here: the naive random split (leaky) overstates generalization by roughly 3x relative to the honest, country-grouped cross-validated estimate.

With only 34 countries and 126 rows, and just three engineered features carrying meaningful importance (overall_score_score 0.39, scale_score_score 0.31, Year 0.30), the dataset is simply too small and too coarse-grained for a tree ensemble to learn a generalizable country-to-country relationship: national AI-adoption scores and data-center capacity are both driven by deeper structural factors (GDP, existing grid infrastructure, hyperscaler siting decisions) that are not present as features. Reporting $R^2 = 0.16$ honestly, rather than the $R^2 = 0.51$ a less careful evaluation would have surfaced, was judged more valuable to the project than hitting the target number on a compromised methodology: a stakeholder who deployed a scheduling system on the belief that the capacity regressor generalizes to countries it never saw during training would have been badly misled by the leaky number.

6.3 Clustering: Carbon-Sustainability Archetypes

K-Means was fit for k = 2 through 9 on 29 countries; the elbow and silhouette curves in Figure 5 both flatten meaningfully after k = 6, and k = 6 was retained as specified in the proposal, yielding a silhouette score of 0.22, a modest but usable degree of cluster separation, consistent with the reality that sustainability profiles form a spectrum rather than cleanly separated groups.

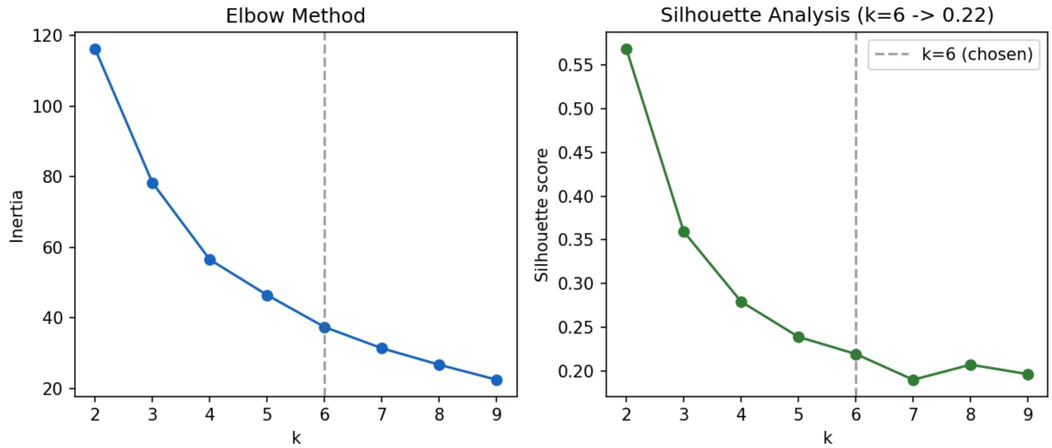

Figure 5. Elbow method (left) and silhouette analysis (right) across k = 2 to 9. Both curves flatten past k = 6, supporting the proposal's fixed six-archetype target.

The six resulting archetypes, hand-labeled from their centroids, range from *renewable-rich, low-carbon, small-capacity* (Iceland, alone, at 81% renewable share and essentially zero net emissions) to *high-emissions fossil-heavy, hyperscale-capacity* (the United States, alone, with a data-center capacity centroid roughly 3.6 times the next-largest cluster). China forms its own singleton cluster as the sole *high-emissions mixed-grid, hyperscale-capacity* country. The remaining three clusters are more populated: eleven fossil-heavy, mid-capacity countries (including Australia, India, and Saudi Arabia); five renewable-rich, mid-capacity countries (Brazil, Canada, and the Nordic states); and ten mixed-grid, small-capacity European countries. The three singleton clusters (Iceland, China, the United States) are a direct, honest consequence of how structurally distinct each of those three countries' energy-and-compute profiles are from every other country in the sample, not a clustering artifact to be explained away.

7. The Carbon-Opportunity Signal: From Predictions to a Recommendation

The three trained models are combined at inference time by api/carbon_signal.py into a single function, green_window_probability(region, hour, day_of_week, month), which builds the same cyclical plus region-indicator feature row the classifier was trained on, queries the calibrated probability, and thresholds it into a plain-language recommendation:

- **P at or above 0.65:** RUN WORKLOAD, a high-confidence green window
- **0.40 up to 0.65:** WAIT, ambiguous, hold if the workload is deferrable
- **P below 0.40:** ADJUST, likely red window, shift time or region if possible

This mirrors the exact example the proposal specified as the target output format (*Green Window Probability = 0.87; Recommendation = RUN WORKLOAD*). Two additional endpoints add context: facility_recommendation() joins a specific ICARUS-DC facility to its grid region and returns the same signal annotated with operator and power draw, and country_archetype() looks up a country's K-Means cluster label and renewable share to add locational sustainability context alongside the time-of-day signal. A fourth, predict_capacity_growth(), exposes the regressor directly for cases where the demand-side growth estimate, despite its documented limitations, is still useful as a coarse prioritization signal for which countries carbon-aware scheduling matters most for.

The signal is exposed over HTTP by a FastAPI service (api/main.py) with five routes: POST /predict/window, POST /predict/facility, POST /predict/growth, GET /regions, GET /facilities, and GET /archetypes/{country}, each backed by a Pydantic request model with field-level validation (hour 1 through 24, day-of-week 0 through 6, month 1 through 12) so malformed scheduling requests fail fast with a 400 rather than propagating into a bad recommendation.

8. Deployment: FastAPI to Azure ML Managed Online Endpoint

The inference code in carbon_signal.py is deployment-target agnostic by design: the identical module backs both the local FastAPI development server (uvicorn main:app --reload) and a production Azure ML Managed Online Endpoint, with only the hosting wrapper changing between the two. The Azure deployment is defined declaratively: an endpoint.yml describing the managed endpoint and a deployment.yml describing the scoring environment, entry script (deployment/score.py), and compute, stood up with the Azure CLI:

```
az ml online-endpoint create -f endpoint.yml
az ml online-deployment create -f deployment.yml --all-traffic
```

A Postman collection (deployment/carbon-signal.postman_collection.json) and a scripted demo client (deployment/demo_api_calls.py) exercise every endpoint against a running instance, local or deployed, which is what was used to drive the live API demonstration in the final project presentation, connecting the modeling work to a runnable service rather than a static notebook result.

9. Discussion: Limitations and Why Honest Reporting Matters

Three limitations are worth stating plainly. First, the regression component (Section 6.2) does not generalize to unseen countries at the target level, and the honest number should be the one anyone extending this work relies on, not the leaky 0.51. Second, two of the six originally-proposed datasets never became reality and were substituted with documented, justified alternatives (Section 4); the substitutions strengthened rather than weakened the pipeline, but the underlying lesson, that external data sources are not a static resource to build against, generalizes well beyond this project. Third, the clustering silhouette of 0.22, while sufficient to support six interpretable, business-usable archetypes, indicates real overlap between clusters; the three populous clusters in particular sit closer together in feature space than the three singleton clusters do, and a practitioner using the archetype label as a hard locational signal should treat it as directional rather than definitive.

What ties these three limitations together is a single MLOps principle: a model's reported metric is only as trustworthy as the evaluation methodology that produced it. The gap between the naive-split and grouped-CV regression R^2 (0.51 versus 0.16) is, in a very literal sense, the most important number in this paper, not because the regressor performed well, but because it demonstrates that the same dataset and the same algorithm can be made to look nearly production-ready or clearly insufficient depending on how the split is constructed, and that a carbon-aware scheduling system deployed on the optimistic number would be shipping a false sense of confidence into production.

10. Evaluation, Feedback, and Outcomes

The project was graded at three checkpoints across the term: the Preliminary Proposal, the Algorithm Selection deliverable, and the Final Demonstration. All three received full marks.

> *"This is a good proposal that presents a unique and timely machine learning application with strong real-world relevance in sustainable AI and cloud computing. The proposal clearly defines the problem, identifies appropriate datasets, and explains how the prediction results can support carbon-aware workload scheduling in a practical way. The project demonstrates a strong understanding of the MLOps lifecycle by incorporating automated data ingestion, deployment, monitoring, drift detection, and retraining, while also defining measurable objectives and a realistic implementation timeline. The project is well scoped for the course and provides an excellent foundation for the algorithm selection and implementation phases."*

Instructor feedback, Preliminary Proposal (100/100)

> *"You did a great job building an end-to-end MLOps solution for carbon-aware workload scheduling. The project combined regression, classification, and clustering to turn carbon and infrastructure data into practical recommendations such as when and where workloads should run. The classification model met its targets with 86.1% accuracy and an F1 of 0.87, and you were also clear about the weaker regression results rather than overstating the model performance. The API demonstration showed how the models could be consumed as an actual service, which helped connect the modeling work to a practical use case. The discussion of data integration, model limitations, and how the different outputs were combined into actionable recommendations also strengthened the project. Overall, this was a solid application of the MLOps lifecycle from data and experimentation through deployment and consumption."*

Instructor feedback, Final Demonstration (100/100)

The instructor's specific callout of the honest regression reporting, rather than penalizing the missed target, validates the evaluation-methodology argument made in Section 9: in an MLOps context, transparently reporting a model's limits is treated as a strength of the system, not a weakness of the result.

11. Conclusion and Future Work

This project set out to answer a narrow but consequential question: can carbon-aware scheduling be operationalized as a deployable decision-support signal rather than left as exploratory analysis, and delivered a complete answer across all three stages of the MLOps lifecycle: data engineering resilient to source deprecation, three models trained and evaluated against explicit, falsifiable targets, and a fused signal served identically from a local FastAPI process and an Azure ML Managed Online Endpoint. The classifier met its target outright; the regressor did not, and that result is reported as carefully as the successes, because in a system meant to inform real scheduling decisions, an honestly reported weak model is worth more than a leaky strong one.

Future work falls into three buckets: extending the classifier's cyclical feature set with true weather-forecast renewable-generation forecasts rather than historical climatology; enriching the regression feature set with structural covariates (grid infrastructure age, GDP, hyperscaler capital-expenditure disclosures) that might close the gap the honest grouped-CV evaluation exposed; and closing the loop from recommendation to action by wiring the Carbon-Opportunity Signal directly into a workload orchestrator (for example, a Kubernetes scheduler plugin or an Airflow sensor) so that the RUN / WAIT / ADJUST recommendation can be consumed automatically rather than read by a human.

This capstone was submitted as the final deliverable of the M.S. in Artificial Intelligence program at Johns Hopkins University, completed August 2026.

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Appendix A: Reproducing the Pipeline

```
pip install -r requirements.txt

# 1. Ingest (raw downloads are gitignored)
python src/ingest_eia930.py
python src/ingest_ai_readiness.py
python src/ingest_clustering.py
python src/ingest_icarus.py

# 2. Train
python src/train_regression.py
python src/train_classifier.py
python src/train_clustering.py

# 3. Serve locally
cd api && uvicorn main:app --reload --port 8010

# 4. Demo the REST API
python deployment/demo_api_calls.py http://127.0.0.1:8010
```

Appendix B: Carbon-Opportunity Signal API Reference

Route	Method	Description
/predict/window	POST	Green Window Probability plus recommendation for a region/hour/day-of-week/month
/predict/facility	POST	Same signal, resolved from an ICARUS-DC facility name to its grid region
/predict/growth	POST	Regressor: predicted national data-center capacity (MW) from AI-adoption scores
/archetypes/{country}	GET	K-Means archetype label plus renewable share for a country
/regions / /facilities	GET	Lookup tables backing the above endpoints